SCIENTIFIC AUDIT

The free-scalar thermal envelope

Conjectures, reproduced calculations, adversarial review and saddle-search limits

Prepared for Grounded DI LLC | 28 October 2026

Disposition: No counterexample or consequential mathematical error was established in this work. Both universal inequalities remain conjectures. The nonconstant competing-saddle calculation remains unresolved.

This report consolidates model-assisted work and executed checks from one working session. It is not an independent expert review or certification.

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1 Executive assessment

Herzog and Stergiou propose that a free scalar bounds the thermal entropy-to-stress-tensor ratio of every unitary three-dimensional conformal field theory (CFT). They also propose a stronger hyperbolic heat-capacity envelope. They present a conditional result near the conformal temperature and test selected examples; they do not claim a universal proof [1].

The audit reproduced the free-field ratios, selected thermal-response values, two planar Chern–Simons benchmarks, finite-N input combinations and several next-to-leading coefficients. A second pass reproduced the recorded numerical output byte-for-byte on the same runtime and narrowed the interpretation of those checks. The sampled calculations use the paper's spectral densities and saddle equations; replay is not a validation of every physical premise.

A subsequent phase-transition search found no qualifying counterexample. A constant-auxiliary-field ordered solution is excluded on the checked reference branch under normalizability assumptions. A coupled nonconstant branch has not been solved or excluded. The final finite-disk work calibrated only one radial Laplacian sector; it did not implement the interacting Green function, determinant, nonlinear saddle search, or free-energy comparison.

Claim map

Claim	Status and scope
$c_S / C_T \leq 16\pi\zeta(3)$	Universal flat-space conjecture; free real scalar saturates.
$\Psi'(x) \leq \Psi'_{\text{scalar}}(x)$, $x \geq 1$	Stronger hyperbolic conjecture; implies the flat-space bound given the stated high-temperature limit.
Near $x = 1+$	Conditional local conclusion using smoothness and modular-boundary assumptions.
Global examples	Free, large-N, holographic and numerical tests support particular cases; no universal proof.

2 Scientific checks

In the paper's conventions, $s(T) = cS\,T^2$ and the separated stress-tensor two-point function is normalized by CT. The conformal temperature is $T0 = 1/(2\pi R)$, $x = T/T0$, and $\Psi = 12R^2\,sH/(\pi^3\,CT)$, where sH is hyperbolic entropy density [1, Eqs. (1), (2), (9)–(13)].

Free fields

For one real scalar, the flat-space thermal free energy is $f = -\zeta(3)\,T^3/(2\pi)$, so $cS = 3\zeta(3)/(2\pi)$. A separated-point Wick calculation gives $CT = 3/(32\pi^2)$. Their ratio is $16\pi\zeta(3) = 60.42197018$. For one two-component complex Dirac fermion, the entropy integral and cited CT convention give $12\pi\zeta(3) = 45.31647763$ [1, Eqs. (3)–(4); 3]. These are free-field checks, not a proof for interacting CFTs.

Conditional local argument

Using the paper's twist-operator inputs and thermodynamic identities gives $\Psi''(1) = 3/4 + t4/240$. For theories to which the boundary-free formula applies, the collider constraint gives $11/15 \le \Psi''(1) \le 23/30$. The conformal scalar requires a modular-Hamiltonian boundary term. The cited correction $\delta h''1 = -\pi/512$ gives $\delta\Psi''\text{scalar}(1) = 1/2$ and $\Psi''\text{scalar}(1) = 19/15$ [1, Eqs. (16)–(24); 4, 5].

For each covered theory, right differentiability and the stated identities give $D(1+\delta) = a\delta + o(\delta)$, where $D = \Psi'\text{scalar} - \Psi'$ and $a \ge 1/2$. Therefore D is positive in a sufficiently small, potentially theory-dependent right neighborhood. A uniform neighborhood is not established. The integrated parity-odd/contact-term treatment was not independently completed.

Selected reference-saddle response

x	Critical O(N), N → ∞	Free scalar
1.1	1.080605	1.126612
2	1.907122	2.250000
5	4.742819	5.870499

These values are Ψ', evaluated from the stated gap and entropy formulas. They establish the comparison at sampled points on the reference saddle, not global phase dominance.

3 Numerical evidence and corrections

Selected endpoint checks

Using the paper's thermal masses and entropy integrals, the critical O(N) and Bardeen–Moshe–Bander leading-large-N entropy fractions reproduce 4/5 and 7/8 of the free-scalar value, respectively. For the planar regular-boson Chern–Simons saddle at $\lambda = 1/2$, the two reported ratios recompute to 49.0337677 and 48.8231125 [1, Eqs. (6), (34), (36), (S43)–(S54)]. These checks assume the physical saddle and cited input equations.

The finite-N Ising and O(N) endpoint rows were recombined from published Casimir and stress-tensor inputs. The O(4) and O(6) rows use stress-tensor lower-bound inputs and are upper estimates. The audit also recombined published coefficients to reproduce the paper's displayed next-to-leading O(N), Gross–Neveu and QED3 expansions. It did not independently redo the source simulations or diagrammatic computations.

Audit of the audit

PASS labels: numerical evaluation of authors' formulas, finite differences sharing a solver, and same-runtime replay establish the specific calculations only. They are not independent validation of the model, boundary conditions, global phase selection, or the universal conjectures.

Local remainder: the positive leading margin applies for each covered theory in a potentially theory-dependent neighborhood. A written $O((x-1)^2)$ term requires suitable remainder control; no uniform neighborhood is established.

Ising: the chosen Monte Carlo amplitude $-0.1531(4)$ and $CT/CT_{,scalar} = 0.946538675(42)$ yield about 51.08. The Monte Carlo paper also reports a thermal-bootstrap-derived comparison $-0.1425(15)$. The latter's conversion, attributed to private communication, was not independently reconstructed. Using the same CT gives about 47.55; both central values are below the conjectured 60.42 bound. The discrepancy is an evidence-quality qualification, not a counterexample [6, 7].

Topological sectors: the handling of global or boundary contributions in the regulated hyperbolic entropy merits an explicit calculation. No inconsistency in Eq. (11) was established; this was downgraded to an unresolved question.

4 Competing phases and saddle search

A first-order transition at $x_c > 1$ with positive entropy-density jump Δs_H would introduce a distributional contribution to the normalized response: $\Psi' \supset [12R^2\,\Delta s_H/(\pi^3\,CT)]\,\delta(x-x_c)$. If the scalar response is finite there and the jump survives the relevant normalization and limits, that would challenge the pointwise envelope. A qualifying unitary theory, replica boundary conditions, physical dominant phases and nonzero jump must all be established.

The literature search did not identify such a case. The earlier critical O(N) analysis reports an instability at or near the conformal point, with the studied instability on the low-temperature side. It did not determine a nonconstant competing saddle's free energy, and could not completely exclude a higher-temperature first-order transition. Holographic low-temperature transitions and non-normalizable-mode examples are not qualifying high-temperature vacuum-Rényi counterexamples without additional checks [2, 8].

Constant auxiliary field

On the checked symmetric $x > 1$ branch, the fundamental fluctuation operator is $K = -\partial\tau^2 - \Delta H^2 - 1/(4R^2) + m^2/R^2$. For normalizable fields, $-\Delta H^2 \geq 1/(4R^2)$, and $m > 0$. Therefore $\langle\varphi, K\varphi\rangle \geq (m^2/R^2)||\varphi||^2$, excluding a nonzero normalizable solution of $K\varphi = 0$ with that constant auxiliary field. This says neither that the complete coupled Hessian is positive nor that a disconnected nonconstant saddle cannot dominate.

A physical competing branch would require solving the coupled equations $K\sigma\,\varphi = 0$ and $\varphi^2 - \sigma + G\sigma(z,z) = 0$ with common regulator, counterterms, contour and replica boundary conditions. No such admissible branch was found or constructed here. No ΔF, entropy jump or transition temperature was calculated.

The strict $N \to \infty$ limit, hyperbolic-volume limit, regulator removal and finite-N limit are distinct. Finite N does not by itself rule out a thermodynamic transition. A future intermediate-temperature failure of the hyperbolic conjecture would not automatically disprove the separate flat-space bound.

5 Radial calibration

A finite hyperbolic disk with radius parameter L, R = 1, regular origin and Dirichlet outer boundary was used to calibrate the static, rotationally symmetric scalar Laplacian. A cell-centered finite-volume discretization was compared against an independent ODE shooting calculation. The Dirichlet disk is a chosen regulator; its full relation to the vacuum Rényi boundary prescription was not established.

L	Cells	Finite volume λ0	Shooting λ0	Difference
3	100	0.953962753	0.953962189	5.63×10^{-7}
3	400	0.953962224	0.953962189	3.53×10^{-8}
6	100	0.447647739	0.447622822	2.49×10^{-5}
6	400	0.447624379	0.447622822	1.56×10^{-6}
9	100	0.344869693	0.344820649	4.90×10^{-5}
9	400	0.344823714	0.344820649	3.07×10^{-6}

Grid refinement improves agreement for this finite eigenproblem. At L = 9, $\lambda_0 - 1/4 \approx 0.094821$; at x = 1.1 the infinite-volume reference mass satisfies $m^2 \approx 0.000589$. Thus the chosen cutoff's spectral shift is much larger than that mass term near x = 1. This is a regulator limitation, not a continuum instability.

Not completed: the finite-disk angular and Matsubara sums, spatially varying interacting Green function, determinant, common counterterms and contour, nonlinear searches from multiple initial profiles, candidate stability checks, and free-energy comparisons. The infinite-volume gap and entropy replay cannot substitute for finite-regulator calibration of those components.

Next discriminating calculation

Implement a common finite-disk Green function and determinant, verify convergence to the symmetric gap and entropy at x = 1.1 and 2, then search admissible nonconstant profiles and compare consistently renormalized free energies. A finite-cutoff stationary point must not be labeled a continuum saddle or dominant phase without the required checks.

6 Evidence and replay ledger

Evidence class	What the record establishes
Paper claims	Universal inequalities are conjectures; local and model-specific results carry stated assumptions.
Executed calculations	Free-field checks, sampled saddle responses, numerical combinations and radial eigenvalue calibration.
Adopted inputs	Paper spectral densities, gap equations, thermal saddles and selected published numerical coefficients.
Inference	Phase transitions could violate the stronger envelope if all stated conditions hold.
Untested	Global nonconstant saddle dominance; universal bounds; independently validated parity-odd contacts.

Reproducibility

The 16-page source PDF (v1, submitted 23 October 2026) has SHA-256:

31693410dcc33967f8bbd1f48833e6c7ac64e49b084674d5d2e4073813f3c563

The v2 O(N) source PDF has SHA-256:

b0e3b679c4766e07acb8453146147a0ab11426f8888f8bf2951e30101a2b421a

Python 3.12.14; NumPy 2.3.5; SciPy 1.17.0; Linux x86-64. Numerical replay on the same runtime matched 136 recorded numeric values and the recorded output bytes. This establishes reproducibility on that runtime, not independent scientific validation.

Radial calibration command: python3 -B radial_calibration.py > radial_results.json

Radial code SHA-256: 5c2f02df065ac5247ddc0e94c85dcd14a7342723bf7e6afe38d769ee547835a4

Radial output SHA-256: cafa6f78c21e095e223d7f3b501106ebfcf09eba2b46066d8e378678215ed6fd

Supporting records

The original audit, symmetric-saddle cross-check and radial calibration packages are retained privately. The listed commands and hashes identify recorded work; this report does not promise public access to the full replay packages. Archived code and outputs were inspected for release, without rerunning calculations. A separate receipt for the reported second-pass byte match was not available.

References and source notes

[1] C. P. Herzog and A. Stergiou, The Free Scalar as a Thermal Envelope for Three-Dimensional CFTs, arXiv:2610.33179v1 (2026). Main text and supplemental material. arXiv record

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[3] L.-Y. Hung, R. C. Myers and M. Smolkin, Twist operators in higher dimensions, arXiv:1407.6429, Eqs. (2.21), (2.45), (2.46), (3.14), (3.23). arXiv record

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[6] A. Bulgarelli, M. Caselle, A. Nada and M. Panero, Casimir effect in critical O(N) models from non-equilibrium Monte Carlo simulations, arXiv:2505.20403, Table II. arXiv record

[7] C.-H. Chang et al., Bootstrapping the 3d Ising stress tensor, arXiv:2411.15300, Table 1. arXiv record

[8] A. Belin, A. Maloney and S. Matsuura, Holographic phases of Rényi entropies, arXiv:1306.2640v4. arXiv record

The authors report AI assistance with literature synthesis, proof attempts, cross-checks, code, figures and revision, with author checking and responsibility for the final manuscript. This report did not inspect the complete model transcripts or author verification notebooks. The arXiv record inspected did not establish journal acceptance or publication [1].

Original research: Christopher P. Herzog and Andreas Stergiou. Audit and synthesis: Grounded DI LLC, with model assistance. Source authorship remains with the cited researchers. No separate researcher independently reviewed or certified this report.